

Name: _____

Class: _____

Date: _____

Worksheet · Section 2.2

Data Compression



The Big Picture

Compression makes files smaller. Practice choosing lossless vs. lossy, encoding and decoding with run-length encoding, and explaining the trade-offs.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

_____ 1. **Data compression**

_____ 2. **Lossless**

_____ 3. **Lossy**

_____ 4. **Run-length encoding**

_____ 5. **Redundancy**

_____ 6. **Decompression**

_____ 7. **Trade-off**

_____ 8. **Approximation**

A. Replacing runs of repeated values with a count and the value

B. Reducing the number of bits needed to represent data

C. Giving up one benefit to gain another, like quality for size

D. A close but not exact version of the original

E. Restoring compressed data so it can be used

F. Repeated patterns in data that compression takes advantage of

G. Compression where the original can be rebuilt exactly

H. Compression that permanently removes some data

Part 2 · Fill It In

Use each word from the word bank once.

exact

redundancy

lossless

approximation

algorithm

lossy

RLE

smaller

1. ZIP files use _____ compression so every byte comes back.
2. Streaming video uses _____ compression to save bandwidth.
3. Long runs of the same pixel color are an example of _____.
4. Lossy compression usually makes files _____ than lossless.
5. Lossless compression lets you rebuild the _____ original.
6. After lossy decompression, you get an _____ of the original.
7. "5A3B" is an example of _____.
8. How much a file shrinks depends on its redundancy and the _____ used.

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Practice

Part 3 · Lossy or Lossless?

Check the best type of compression for each file.

Item	Lossy	Lossless
1. Bank statements		
2. Music on a streaming app		
3. Python source code		
4. A live video call		
5. Medical MRI scans		
6. Photos posted to social media		
7. A novel's manuscript		
8. Thumbnail preview images		

Part 4 · Encode & Decode (RLE)

Encode WWWWWB BBBWWWWBBB → _____

Encode AABBBBBBCD → _____

Decode 3X1Y4Z → _____

Decode 1A1B1C → _____

“ABC” is 3 characters; its RLE version is _____ characters. RLE made it _____!

Part 5 · Pixel Grid

Encode each row of this 5 × 5 image (B = black, W = white) with RLE.

Row	Pixels	RLE
1	BBBBB	_____
2	BWWWB	_____
3	BWBWB	_____
4	BWWWB	_____
5	BBBBB	_____

Total RLE characters: _____ vs. 25 original. Did RLE help? _____

Part 6 · Make the Call

A museum is digitizing 10,000 paintings for (1) permanent archives and (2) an online gallery. Recommend a compression type for each use and explain.

**Think:**

Ask: “Do I need it back EXACTLY?” Yes → lossless. “Is small and fast more important than perfect?” → lossy.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What does lossy compression do?
- A. Permanently removes some data to shrink the file
 - B. Encrypts the file
 - C. Keeps every bit of the original
 - D. Always makes the file bigger

★★ On Level

2. Which RLE string represents GGGGRRB ?
- A. 4G1R2B
 - B. 4G2R1B
 - C. G4R2B1
 - D. 7GRB

★★ On Level

3. Which file is likely to shrink the MOST with lossless compression?
- A. An already-compressed JPEG
 - B. A log file where the same lines repeat many times
 - C. An encrypted file
 - D. A file of random numbers

★★★ Exam Hard

4. A photo is saved with lossy compression, opened, and saved again with lossy compression many times. What is most likely?
- A. The quality may get worse each time
 - B. The file size doubles each time
 - C. The quality stays exactly the same
 - D. The file becomes lossless

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)
- A. Lossless compression can reduce size without losing information.
 - B. Lossy compression is always the better choice.
 - C. Run-length encoding always reduces file size.
 - D. The right choice depends on whether exact reconstruction is needed.

Part 8 · Written Response

Explain why a 1-hour movie streams smoothly on a phone, but a lossless copy of the same movie might not.
